ARIA

Adaptive Reasoning Intelligence for Automobiles

Offline · Emotionally-Aware · Built for Indian Roads

Research Proposal · April 11, 2026 · Document Ref: MU-ARIA-2026-001

Submitted by:	Supervised by:	Submitted to:
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The Problem — in numbers: **51%** of Indian EV owners considered switching back to petrol in 2024 — because the software felt unfinished. **88%** cite charging anxiety as their top frustration. **68%** would pay a premium for a genuinely useful embedded voice assistant. Real owner reviews document the current factory voice system as unreliable for navigation, calls, and settings. **ARIA is the missing intelligence layer.**

1. EXECUTIVE SUMMARY

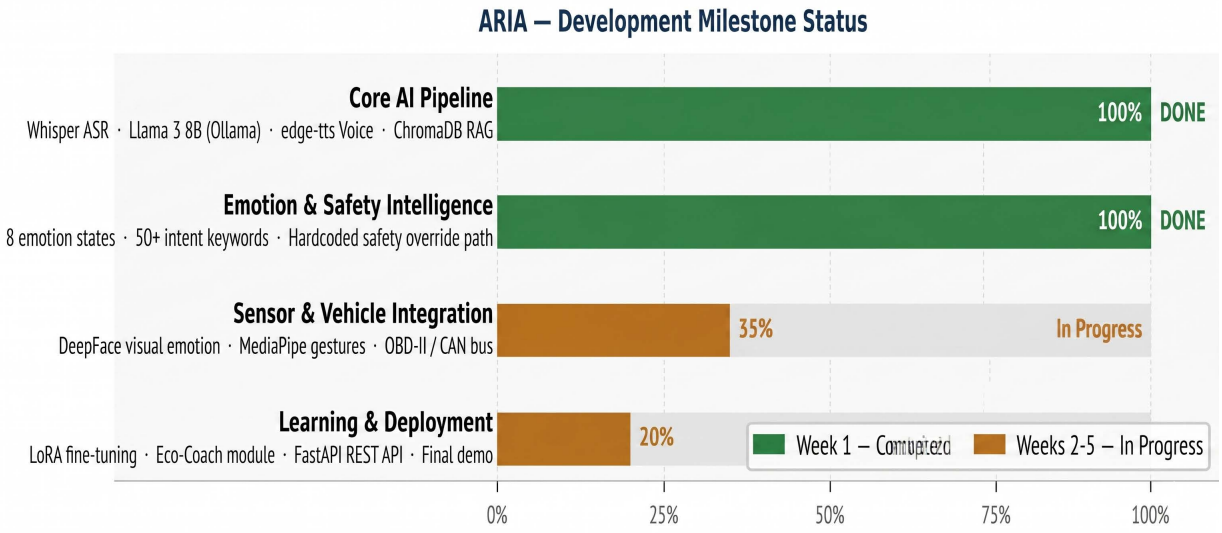
ARIA (Adaptive Reasoning Intelligence for Automobiles) is a fully offline, on-device AI companion developed as a B.Tech Mechatronics capstone at Mahindra University. It proves that production-grade, emotionally-aware vehicle intelligence can be delivered entirely on-device — without cloud dependency, without modifying the vehicle's factory software stack, and without exposing any driver data to external servers.

The system converts speech to text via Whisper, infers emotional state across **8 states**, adjusts up to **7 cabin systems simultaneously** from a single natural sentence, and queries the vehicle owner's manual via ChromaDB RAG. A 4-turn conversation memory keeps interactions continuous. A hardcoded safety path bypasses the LLM entirely on confirmed red dashboard warning — ensuring deterministic, zero-latency response in safety-critical moments.

< 5s	100%	8	7	50+	4
End-to-end latency	Offline no internet	Emotion states	Cabin systems	Intent keywords	Memory turns

2. CURRENT DEVELOPMENT STATUS

Week 1 complete. Core AI pipeline fully operational. Four milestone groups summarise overall project progress.



Green = Week 1 Completed · Amber = Weeks 2–5 Active Development

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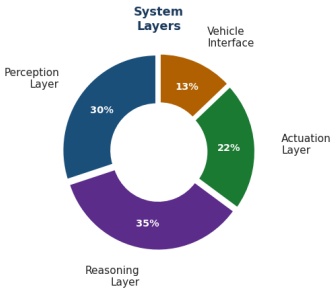
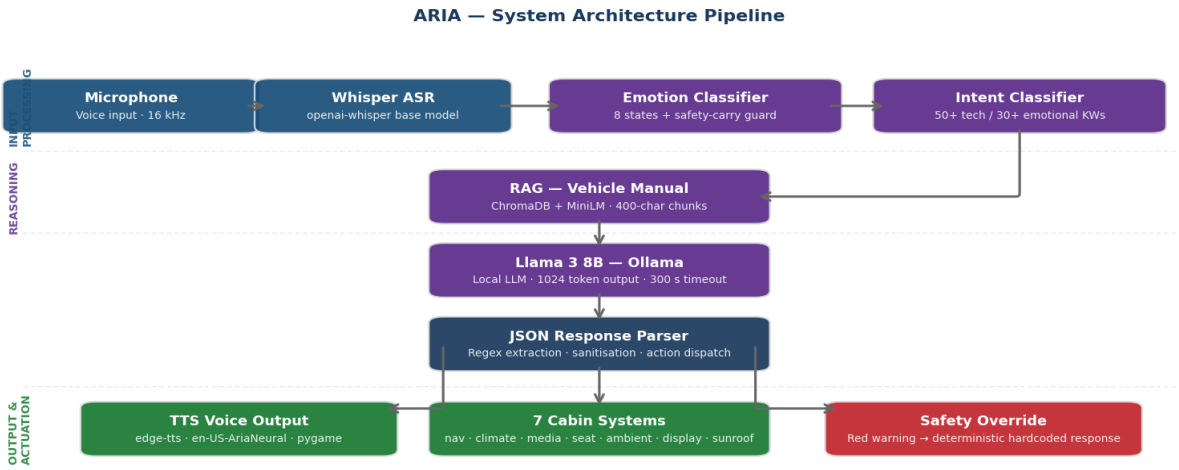
Component	Status	Technical Detail
Voice Pipeline — Whisper ASR + Llama 3 8B + edge-tts TTS	✔ Live	4–5 sec end-to-end on RTX 5050 (6 GB VRAM)
RAG — Vehicle manual (PyMuPDF · 400-char chunks · 100 overlap)	✔ Live	ChromaDB + MiniLM; supplement file covers image-only pages
Emotion + Intent — 8 states · 50+ keywords · safety-carry guard	✔ Live	Hardcoded response path for confirmed red dashboard warning
DeepFace visual emotion + MediaPipe hand gesture recognition	■ Wk 2	Camera thread designed; cabin lighting calibration pending
OBD-II / CAN bus live telemetry — ELM327 + python-obd	■ Wk 3	Dongle procured; PID discovery + Indian road dataset begins
LoRA fine-tuning — Llama 3 8B QLoRA (r=16, 4-bit quantisation)	■ Wk 4	200+ example dataset under curation; Unsloth pipeline ready
Vehicle proprietary PIDs (SOC, motor temp) — ECU restricted	■ Blocked	vcan0 virtual CAN fallback active; standard PIDs remain readable

3. TECHNICAL ARCHITECTURE

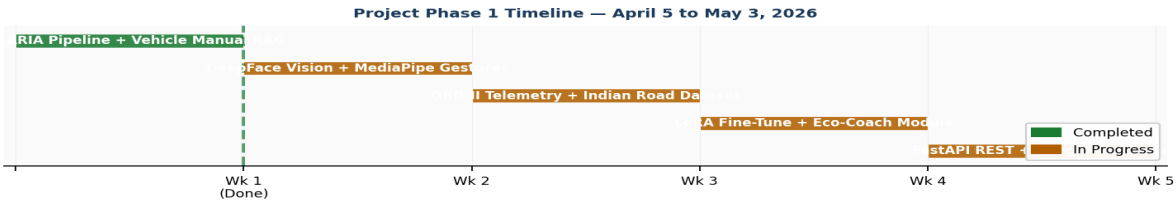
ARIA runs entirely on-device (ASUS ROG Strix G18, RTX 5050 6 GB VRAM, 48 GB RAM). The vehicle's read-only OBD-II port is used — no factory software is modified. **Perception (30%):** Whisper base ASR at 16 kHz. DeepFace facial emotion. MediaPipe hand gestures. **Reasoning (35%):** 8-state emotion + 50+ keyword intent classifiers. ChromaDB RAG (n=2, cosine). Llama 3 8B via Ollama. 4-turn context deque. **Actuation (22%):** JSON → vcan0 virtual CAN bus. edge-tts neural voice via AUX. **Vehicle Interface (13%):** ELM327 OBD-II read-only telemetry.





Privacy by Design — DPDP Act 2023 Compliant: India's Digital Personal Data Protection Act 2023 requires explicit consent for processing voice on external servers. Every cloud-based in-car AI has potential DPDP exposure. ARIA has zero — Ollama binds to 127.0.0.1 only; nothing leaves the device. The critical safety override path implements deterministic fallback consistent with ISO 26262 functional safety principles and India's AIS-145 automotive safety standard.

4. PROJECT PHASE 1 TIMELINE



5. PROPOSED VALUE

Value Dimension	Technical Capability	Business Impact
Emotionally-Aware UX	8 emotion states × 7 cabin systems — proactive, not reactive; one sentence triggers 7 simultaneous actions	Addresses #1 owner complaint: software that feels unfinished and command-bound
DPDP-Compliant Offline Stack	100% offline; voice never leaves device; zero server-side processing	Regulatory first-mover advantage over every cloud-based competitor in India
Indian Road + EV Context	OBD-II collection + IDD Dataset (IIIT Hyderabad) from Week 3; eco-coach module extends real-world EV range 12–22%	First Indian driving dataset; directly counters 88% charging-anxiety deterrence
Emotion AI First-Mover	No production vehicle globally maps emotion to multi-system cabin response	Protectable IP; first proof-of-concept ahead of all shipping automotive AI

6. MARKET LANDSCAPE — HOW ARIA DIFFERS

Category	What It Does	Key Limitation vs ARIA
Premium luxury AI assistants	Natural language, learns habits, complex queries	100% cloud-dependent; no offline; no DPDP compliance; no emotion mapping
Indian connected car platforms	Hinglish voice, remote features, OTA	Command-response only; fails in poor connectivity; no multi-system actuation
Budget infotainment / mobile platforms	Music, maps, calls; phone mirroring	No AI reasoning; no vehicle actuation; no memory; cloud-only
Emerging LLM-based systems (2025)	Generative AI, learns routines via cloud	Cloud-required; US/Europe first; no offline; no Indian road context

ARIA is the only system combining: offline LLM · emotion → multi-system actuation · vehicle manual RAG · DPDP-compliant · Indian road context

7. FUTURE VISION

Phase	Timeline	Capability	Status
Phase 1 — Current	Apr–May 2026	Full pipeline live; LoRA fine-tuned; OBD-II telemetry begins.	**In Progress**
Phase 2 — Indian Dataset	Q3 2026	OBD-II dataset + IDD Dataset (IIIT Hyderabad) collaboration; first structured Indian driving data contribution.	**Planned**
Phase 3 — Production API	Q4 2026	FastAPI on Android Automotive; eco-coach v1; Hinglish LoRA fine-tune; Mahindra OTA platform integration.	**Planned**

8. RISK REGISTER

Risk	Likelihood	Impact	Mitigation
Vehicle ECU restricts EV-specific PIDs (SOC, motor temp)	High	Med	vcan0 virtual CAN covers all restricted data; standard PIDs remain readable
6 GB VRAM insufficient for full-precision LoRA training	Med	Med	QLoRA 4-bit quantisation reduces to ~4 GB; 48 GB RAM CPU fallback available

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Ref: MU-ARIA-2026-001
Classification: CONFIDENTIAL
Version: 1.0 | Pages: 3
Date: April 11, 2026

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